

SIMPLIFIED AIRCRAFT TAKEOFF PERFORMANCE ANALYZER

(PYTHON-BASED PROJECT)

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INTRODUCTION

Aircraft takeoff performance is influenced by several aircraft, aerodynamic and runway conditions. Parameters such as aircraft mass, engine thrust, wing area, takeoff speed, air density and runway length affect the basic performance calculations used during the takeoff phase.

To study these relationships, this project develops the Simplified Aircraft Takeoff Performance Analyzer, a Python-based educational program. The program accepts aircraft and runway information and performs elementary calculations including dynamic pressure, lift, drag, net force, acceleration and required takeoff distance.

The calculated required runway distance is then compared with the available runway. The program calculates a runway safety margin and classifies the result as SUFFICIENT RUNWAY or INSUFFICIENT RUNWAY.

The project demonstrates how Python programming concepts can be applied to an aerospace engineering problem. It is intended as an educational model and not as an operational aircraft performance or safety system.

PROBLEM STATEMENT

Aircraft takeoff performance is affected by aircraft and environmental conditions such as mass, engine thrust, wing area, aerodynamic coefficients, takeoff speed, air density and runway length.

The main issue addressed by this project is the development of a simple Python program which uses selected aircraft and runway parameters to estimate the acceleration and runway distance required for takeoff, and then evaluates whether the available runway is sufficient.

OBJECTIVE

- For creating a takeoff performance analyzer using Python.
- For accepting aircraft, environmental and runway inputs from the user.
- For calculating dynamic pressure, lift and drag.
- For calculating net force and aircraft acceleration.
- For calculating the required takeoff distance.
- For calculating the runway safety margin.
- For categorizing the runway condition as SUFFICIENT RUNWAY or UNSUFFICIENT.

FUNCTIONAL REQUIREMENTS

The functional requirements list the primary functions of the Simplified Aircraft Takeoff Performance Analyzer.

User Input

The program accepts the following aircraft values from the user:

- Aircraft mass
- Engine thrust
- Wing area
- Lift coefficient
- Drag coefficient
- Takeoff speed

The program also accepts environmental and runway values:

- Air density
- Wind speed
- Runway length
- Runway slope

Input Validation

Basic validation is included for aircraft mass, and the takeoff-distance calculation checks that acceleration is greater than zero.

Calculation Outputs

- Dynamic pressure
- Lift
- Drag
- Net force
- Acceleration
- Required takeoff distance
- Runway safety margin
- Runway assessment

CALCULATION REQUIREMENTS

The program performs the main aerodynamic and takeoff calculations through functions contained in the calculation modules.

Dynamic Pressure

Dynamic pressure is calculated using $q = 0.5 \times \rho \times$

V^2 . Lift and Drag

The program calculates lift and drag using the dynamic pressure, wing area and the supplied aerodynamic coefficients.

Net force and acceleration

Net force is calculated as engine thrust minus drag. Acceleration is calculated by net force divided by aircraft mass.

Takeoff Distance

The simplified takeoff-distance model uses $s = V^2 / (2a)$, where V is takeoff velocity and a is acceleration.

NON-FUNCTIONAL **REQUIREMENTS**

These non-functional requirements describe how the project should operate and its intended characteristics.

Usability

The interface should remain straightforward and user-friendly. The user enters the required aircraft and runway information and views the calculated output through the console.

Reliability

The same inputs should produce consistent calculations. Basic validation is included to reduce errors caused by invalid values.

Performance

The program performs mathematical calculations locally, so the calculations are intended to complete quickly without external services.

Maintainability

The project is divided into separate Python modules so that input handling, calculations, runway analysis and reporting can be maintained independently.

ERROR HANDLING

The program checks for invalid conditions during input and calculation. For example, aircraft mass is required to be positive, and the takeoff-distance function raises an error when acceleration is not greater than zero.

EDUCATIONAL USE

The design emphasizes simple calculations and modular Python code so that aerospace engineering students can understand the relationship between programming logic and basic aircraft performance calculations.

RESOURCE EFFICIENCY

The program runs locally and does not require a database, external API or large external dataset.

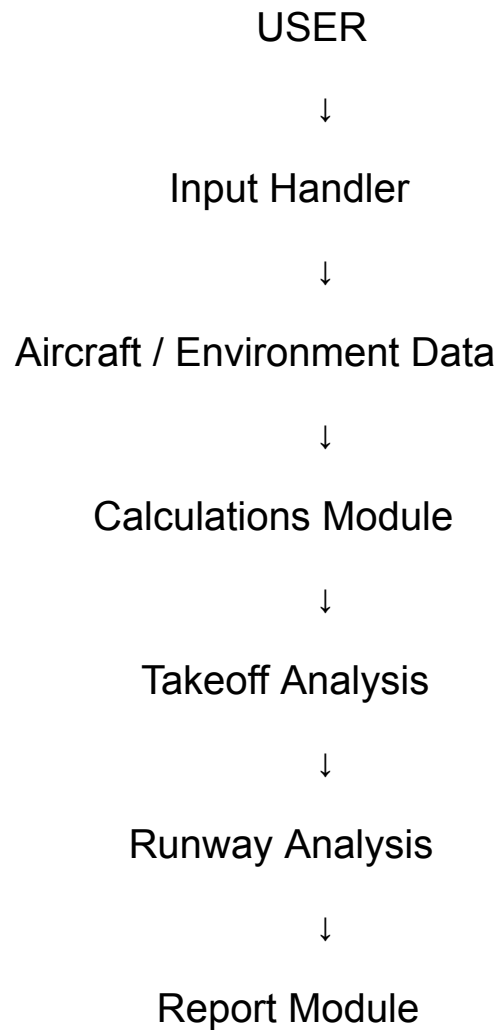
PROJECT LIMITATION

The model is intentionally simplified. The current main calculation flow collects wind speed and runway slope but does not incorporate them into the displayed takeoff-distance calculation.

SYSTEM ARCHITECTURE

The Simplified Aircraft Takeoff Performance Analyzer uses a modular sequential structure. The user enters aircraft and environmental/runway data, the program performs aerodynamic and force calculations, estimates the required runway distance, analyzes the runway and displays the final report.

Arithmetic flow

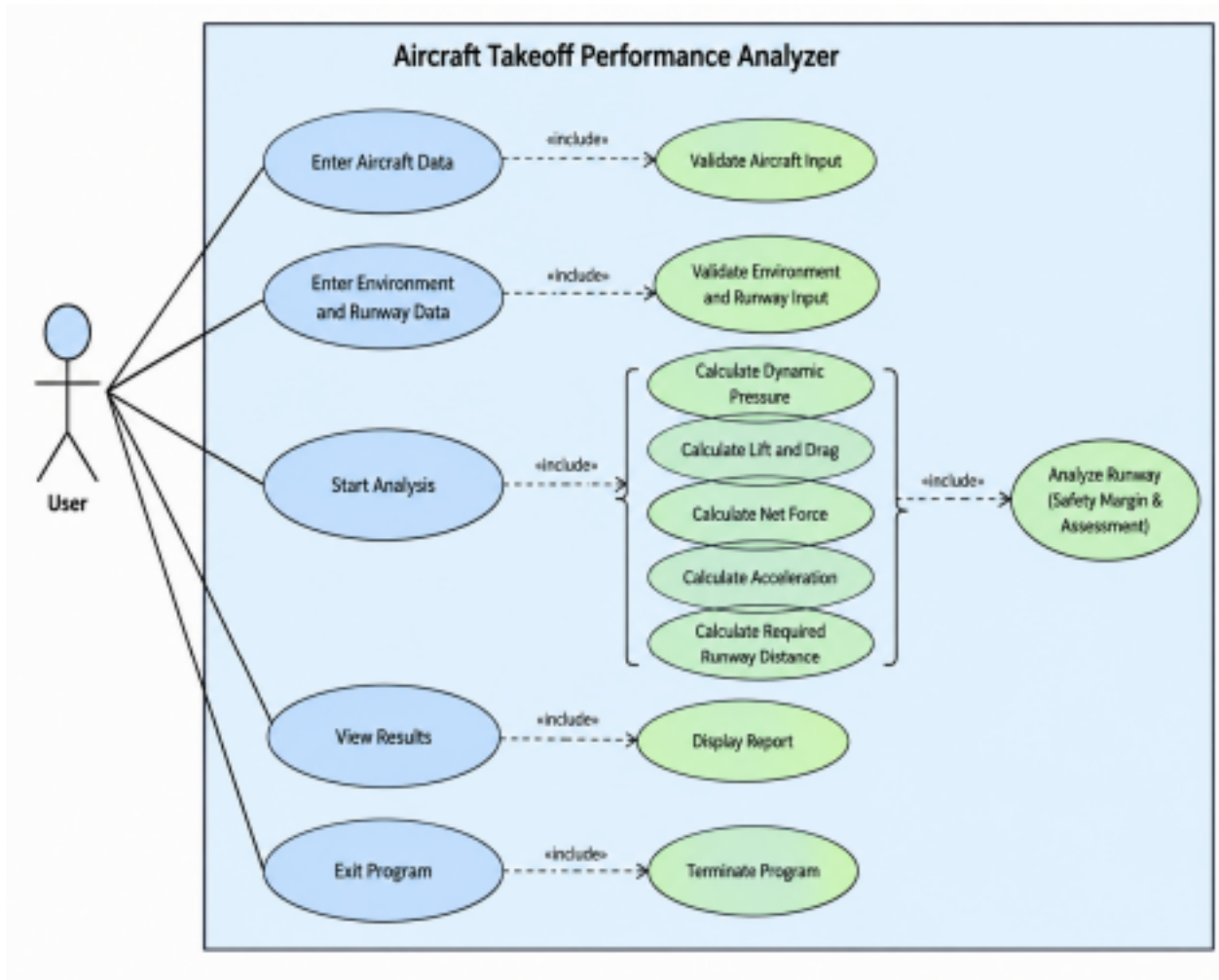


The flow represents the sequence followed by the current modular Python project.

DESIGN DIAGRAM

USE CASE DIAGRAM

The use case view represents the interaction between the user and the aircraft takeoff performance analyzer.



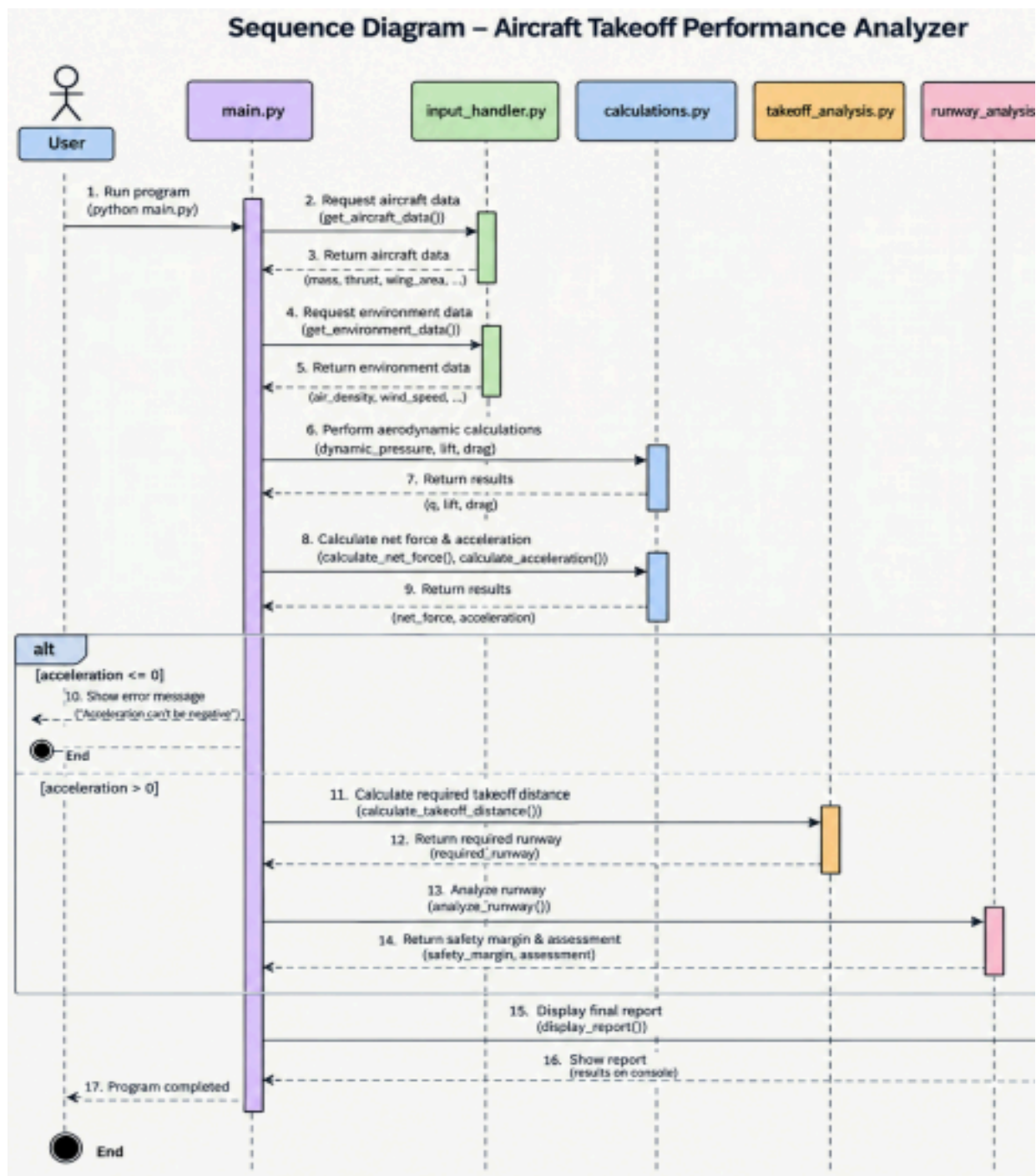
The system receives the required inputs and returns the calculated takeoff and runway results.

WORKFLOW DIAGRAM

The workflow represents the program's steps from entering data to the final runway assessment

SEQUENCE DIAGRAM

The sequence diagram represents the interaction between the user and the program's logical stages.



CLASS/COMPONENT DIAGRAM

The system is designed using a modular component-based structure, where each Python file performs a specific task.

Module Main- responsibility
main.py -Central controller for the complete analysis
input_handler.py - Collects aircraft and environment/runway inputs

calculations.py - Dynamic pressure, lift, drag, net force and acceleration
takeoff_analysis.py - Required takeoff distance
runway_analysis.py .- Safety margin and runway assessment
validation.py -Basic positive-value validation
constants.py -Stores the gravitational constant
report.py - Displays the final results

This modular approach makes the program easier to understand, test, analyze and modify.

DESIGN DECISIONS AND RATIONALE

DECISION	REASON
Command-Line Interface	A command-line interface keeps the project focused on calculations and Python logic
Python	Python provides simple syntax and is suitable for mathematical and engineering calculations.
Modular Structure	Separate modules make individual parts of the project easier to understand and maintain
Parameter-Based Estimation	The program uses user-supplied parameters so it remains self-contained and suitable for an
Simplified Takeoff Model	The project uses basic force and kinematic relationships to demonstrate takeoff performance

Basic Input Validation	Validation helps prevent simple invalid values from being passed into calculations
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IMPLEMENTATION DETAILS

The project is implemented in Python 3 and is designed to run locally through Command Prompt, PowerShell or an IDE such as VS Code.

INPUTS

The input handler collects aircraft mass, engine thrust, wing area, lift coefficient, drag coefficient and takeoff speed. It also collects air density, wind speed, runway length and runway slope.

AERODYNAMIC CALCULATIONS

The calculations module determines dynamic pressure and then uses it with the supplied aerodynamic values to calculate lift and drag.

FORCE AND ACCELERATION

Net force is obtained from thrust minus drag. The resulting acceleration is calculated by dividing net force by aircraft mass.

TAKEOFF DISTANCE

The takeoff-analysis module calculates the required runway distance using the takeoff speed and acceleration.

RUNWAY ANALYSIS

The runway-analysis module subtracts required runway distance from available runway length to obtain the safety margin and then classifies the runway condition.

INPUTS AND CALCULATIONS

Aircraft inputs

- Mass (kg)
- Thrust (N)
- Wing area (m²)
- Lift coefficient (CL)
- Drag coefficient
- Takeoff speed (m/s)

Environment and runway inputs

- Air density (kg/m³)
- Wind speed (m/s)
- Runway length (m)
- Runway slope (%)

Main equations used

Dynamic pressure: $q = 0.5\rho V^2$

Net force: $F_{\text{net}} = T - D$

Acceleration: $a = F_{\text{net}} / m$

Takeoff distance: $s = V^2 / (2a)$

Safety margin: Available runway – Required runway

SCREENSHOTS AND RESULTS

The program produces a console report containing aircraft data, aerodynamic results, takeoff performance and runway analysis.

CATEGORY	RESULT
AIRCRAFT MASS	1111kg
TAKEOFF SPEED	31 m/s
ACCELERATION	1.87085 m/s^2
REQUIRED RUNWAY	256.83 m
AVAILABLE RUNWAY	1800 m
SAFETY MARGIN	1543.17 m
FINAL ASSESSMENT	SUFFICIENT RUNWAY

```

Windows PowerShell
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PS C:\Users\NITRO V15\Desktop\Aircraft_Takeoff_Analyzer> python main.py
=====
AIRCRAFT TAKEOFF PERFORMANCE
ANALYZER
=====

Enter Aircraft information
-----
Enter the mass of the aircraft (kg) : 1111
Enter the thrust produced by the engine (N) : 2400
Enter wing area (m^2) : 16.2
Enter lift coefficient (CL) : 1.5
Enter drag coefficient (DL) : 0.035
Enter flight take off speed (m/s) : 31

Enter Environment and Runway Information
-----
Enter air density (kg/m^3) : 1.18
Enter wind speed (m/s) : 5
Enter runway length (m) : 1800
Enter runway slope (%) : 1.2

=====
AIRCRAFT TAKEOFF PERFORMANCE
ANALYZER
=====

AIRCRAFT DATA
-----
Aircraft mass      : 1111.0 kg
Engine thrust      : 2400.0 N
Takeoff speed      : 31.0 m/s

AERODYNAMIC RESULTS
=====
Dynamic pressure   : 566.99 pa
Lift                : 13777.857 N
Drag               : 321.48333 N

```

```
Windows PowerShell
=====
AIRCRAFT TAKEOFF PERFORMANCE
ANALYZER
=====

AIRCRAFT DATA
-----
Aircraft mass      : 1111.0 kg
Engine thrust      : 2400.0 N
Takeoff speed      : 31.0 m/s

AERODYNAMIC RESULTS
-----
Dynamic pressure   : 566.90 pa
Lift                : 13777.857 N
Drag               : 321.48333 N

TAKEOFF PERFORMANCE
-----
Acceleration       : 1.8708520882088209 m/s^2
Required runway     : 256.8348417431745 m

RUNWAY ANALYSIS
-----
Available Runway    : 1800.0 m
Safety Margin       : 1543.1651582568256 m
Assessment          : SUFFICIENT RUNWAY
=====
PS C:\Users\NITRO V15\Desktop\Aircraft_Takeoff_Analyzer>
```

CATEGORY	RESULT
AIRCRAFT MASS	78000 kg
TAKEOFF SPEED	72 m/s
ACCELERATION	2.902 m/s^2
REQUIRED RUNWAY	893.1326 m
AVAILABLE RUNWAY	4000 m
SAFETY MARGIN	3106.867 m
FINAL ASSESSMENT	SUFFICIENT RUNWAY

```
Windows PowerShell

=====
AIRCRAFT TAKEOFF PERFORMANCE
ANALYZER
=====

Enter Aircraft information
=====
Enter the mass of the aircraft (kg) : 78000
Enter the thrust produced by the engine (N) : 240000
Enter wing area (m^2) : 122.6
Enter lift coefficient (CL) : 1.6
Enter drag coefficient (DL) : 0.033
Enter flight take off speed (m/s) : 72

Enter Environment and Runway Information
=====
Enter air density (kg/m^3) : 1.3
Enter wind speed (m/s) : 11
Enter runway length (m) : 4000
Enter runway slope (%) : 2

=====
AIRCRAFT TAKEOFF PERFORMANCE
ANALYZER
=====

AIRCRAFT DATA
=====
Aircraft mass      : 78000.0 kg
Engine thrust      : 240000.0 N
Takeoff speed      : 72.0 m/s

AERODYNAMIC RESULTS
=====
Dynamic pressure   : 3369.6 pa
Lift               : 660980.736 N
drag              : 13632.72768 N

TAKEOFF PERFORMANCE
=====

31°C Sunny
```

```
Windows PowerShell

Enter lift coefficient (CL) : 1.6
Enter drag coefficient (DL) : 0.033
Enter flight take off speed (m/s) : 72

Enter Environment and Runway Information
=====
Enter air density (kg/m^3) : 1.3
Enter wind speed (m/s) : 11
Enter runway length (m) : 4000
Enter runway slope (%) : 2

=====
AIRCRAFT TAKEOFF PERFORMANCE
ANALYZER
=====

AIRCRAFT DATA
=====
Aircraft mass      : 78000.0 kg
Engine thrust      : 240000.0 N
Takeoff speed      : 72.0 m/s

AERODYNAMIC RESULTS
=====
Dynamic pressure   : 3369.6 pa
Lift               : 660980.736 N
drag              : 13632.72768 N

TAKEOFF PERFORMANCE
=====
Acceleration       : 2.9021445169230766 m/s^2
Required runway     : 893.1326420464067 m

RUNWAY ANALYSIS
=====
Available Runway    : 4000.0 m
Safety Margin       : 3106.8673579535935 m
Assessment          : SUFFICIENT RUNWAY

PS C:\Users\NITRO V15\Desktop\Aircraft_Takeoff_Analyzer> |

31°C Sunny
```

CATEGORY	RESULT
AIRCRAFT MASS	78000 kg
TAKEOFF SPEED	72 m/s
ACCELERATION	2.902 m/s^2
REQUIRED RUNWAY	893.1326 m
AVAILABLE RUNWAY	4000 m
SAFETY MARGIN	3106.867 m
FINAL ASSESSMENT	SUFFICIENT RUNWAY

```

Windows PowerShell
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PS C:\Users\NITRO V15\Desktop\Aircraft_Takeoff_Analyzer> python main.py
=====
AIRCRAFT TAKEOFF PERFORMANCE
ANALYZER
=====

Enter Aircraft information
-----
Enter the mass of the aircraft (kg) : 79010
Enter the thrust produced by the engine (N) : 242000
Enter wing area (m^2) : 124.6
Enter lift coefficient (CL) : 1.7
Enter drag coefficient (DL) : 0.035
Enter flight take off speed (m/s) : 77

Enter Environment and Runway Information
-----
Enter air density (kg/m^3) : 1.1
Enter wind speed (m/s) : 10
Enter runway length (m) : 3000
Enter runway slope (%) : 0

=====
AIRCRAFT TAKEOFF PERFORMANCE
ANALYZER
=====

AIRCRAFT DATA
-----
Aircraft mass      : 79010.0 kg
Engine thrust      : 242000.0 N
Takeoff speed      : 77.0 m/s

AERODYNAMIC RESULTS
-----
Dynamic pressure   : 3260.9500000000003 pa
Lift               : 690734.429 N

=====

```

```

Enter lift coefficient (CL) : 1.7
Enter drag coefficient (DL) : 0.035
Enter flight take off speed (m/s) : 77

Enter Environment and Runway Information
-----
Enter air density (kg/m^3) : 1.1
Enter wind speed (m/s) : 10
Enter runway length (m) : 3000
Enter runway slope (%) : 0

=====
AIRCRAFT TAKEOFF PERFORMANCE
ANALYZER
=====

AIRCRAFT DATA
-----
Aircraft mass      : 79010.0 kg
Engine thrust      : 242000.0 N
Takeoff speed      : 77.0 m/s

AERODYNAMIC RESULTS
-----
Dynamic pressure   : 3260.9500000000003 pa
Lift               : 690734.429 N
Drag              : 14221.002950000002 N

TAKEOFF PERFORMANCE
-----
Acceleration      : 2.8829135179091256 m/s^2
Required runway    : 1028.3000102445135 m

RUNWAY ANALYSIS
-----
Available Runway   : 3000.0 m
Safety Margin      : 1971.6999897554865 m
Assessment         : SUFFICIENT RUNWAY

=====
PS C:\Users\NITRO V15\Desktop\Aircraft_Takeoff_Analyzer>

```

A typical output structure is shown below:

SECTION	RESULTS DISPLAYED
Aircraft Data	Mass, engine thrust, takeoff speed
Aerodynamic Results	Dynamic pressure, lift, drag
Takeoff Performance	Acceleration, required runway
Runway Analysis	Available runway, safety margin, assessment

The final assessment is either SUFFICIENT RUNWAY or INSUFFICIENT RUNWAY based on the calculated safety margin.

TESTING APPROACH

Testing should be performed by executing the program with different combinations of aircraft and runway inputs. The calculations can be checked by comparing the displayed values with the equations used by each module.

The project should also be checked for invalid values, especially non-positive aircraft mass and non-positive acceleration before calculating takeoff distance.

TEST CASES

TEST ID	PURPOSE	EXPECTED BEHAVIOUR
T01	Normal positive inputs	Calculate and display takeoff results
T02	Insufficient runway	Display INSUFFICIENT RUNWAY
T03	Sufficient runway	Display SUFFICIENT RUNWAY
T04	Non-positive acceleration	Reject takeoff-distance calculation
T05	Non-positive mass	Reject invalid aircraft mass

VALIDATION TEST

TEST ID	CONDITION	EXPECTED BEHAVIOUR
VT01	Zero or negative aircraft mass	Display an error and stop
VT02	Zero or negative acceleration	Raise an error for takeoff-distance calculation
VT03	Positive valid inputs	Continue with calculations
VT04	Available runway less than	Display INSUFFICIENT

	required runway	RUNWAY
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These tests focus on the basic validation and decision points present in Current implementation.

CHALLENGES FACED

UNDERSTANDING THE AERODYNAMIC CALCULATIONS

Understanding how air density, velocity, wing area and aerodynamic coefficients are connected to dynamic pressure, lift and drag can be challenging when implementing the formulas in Python.

FORCE AND ACCELERATION

The project requires the aerodynamic drag result to be connected with engine thrust to obtain net force and then acceleration.

TAKEOFF DISTANCE

The takeoff-distance calculation requires a positive acceleration. Handling a zero or negative acceleration condition is therefore an important part of the implementation.

RUNWAY ASSESSMENT

The calculated required runway must be compared with the available runway to produce a meaningful safety margin and assessment.

INPUT VALIDATION

Basic checks are required so that invalid values such as non-positive mass do not proceed through the calculations.

TESTING

Different combinations of aircraft and runway values can be used to verify the calculation flow and runway assessment.

LEARNING AND KEY TAKEAWAYS

This project demonstrates the connection between Python programming and an aerospace engineering application. It also provides practice in organizing a calculation program into multiple modules.

- Using `input()` and converting user input into numerical values.
- Working with variables, constants and arithmetic operations.
- Using functions and importing functions between modules.
- Using if statements and comparison operators.
- Calculating dynamic pressure, lift and drag.
- Calculating net force and acceleration.
- Calculating a simplified takeoff distance.
- Calculating runway safety margin and assessment.
- Adding basic input validation.
- Organizing a Python project into separate modules.
- Preparing technical documentation for the project.

FUTURE ENHANCEMENTS

- Incorporate wind speed into the effective takeoff airspeed calculation.
- Incorporate runway slope into the force and acceleration model.
- Use gravitational acceleration and aircraft weight explicitly in an expanded runway model.
- Improve the aerodynamic lift calculation and parameter naming.
- Add comprehensive validation for every user input.
- Add automated unit tests for each module.
- Provide graphs showing acceleration and runway-distance relationships.

- Provide a graphical user interface.
- Allow aircraft data to be stored and reused.
 - Expand the model with additional atmospheric and aircraft-performance parameters.

REFERENCE

- 1) Python 3 Documentation – Python programming language reference.
- 2) Project Source Code – main.py, Aircraft Takeoff Performance Analyzer.
- 3) Project Source Code – calculations.py, input_handler.py, takeoff_analysis.py, runway_analysis.py, validation.py, constants.py and report.py.
- 4) Project Statement – statement.md.
- 5) Project README – README.md.